

Introduction to probability

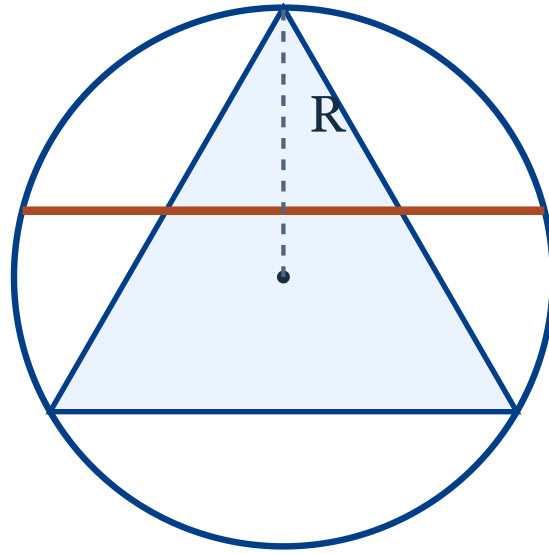
Supplementary · Bertrand's paradox and sampling rules

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Random selection of a chord

A chord is selected in a circle. What must be specified before its length can be assigned a probability?



Triangle side = $\sqrt{3} R$

Chord length = L

What is $P(L > \sqrt{3} R)$?

We need a rule for selecting the chord. Later we will calculate the answer for three different rules.

The chord-length question

For a circle of radius R , find the probability that a sampled chord has length $L > \sqrt{3}R$. What sampling rule is missing from the question?

The reference length is $\sqrt{3} R$.

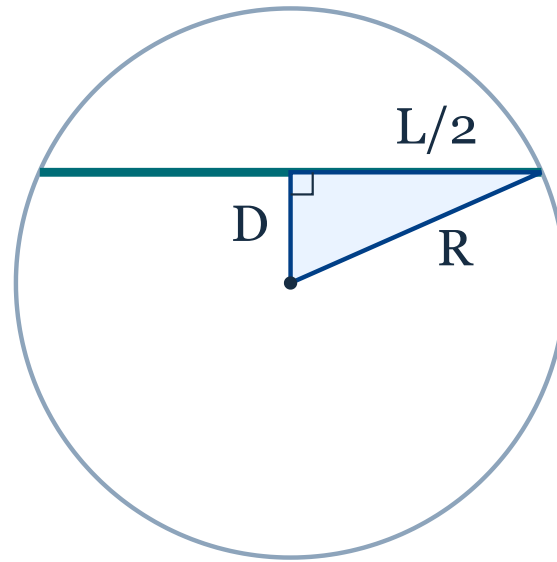
We will calculate the probability under three specified sampling rules.

In each case the required event is the same geometric condition:

$$L > \sqrt{3} R.$$

Chord length from the midpoint distance

Let D be the distance from the centre to the chord. Express its length L in terms of D .



$$(L/2)^2 + D^2 = R^2$$

$$\text{Long} \Leftrightarrow D < R/2$$

The right triangle gives $(L/2)^2 + D^2 = R^2$.

Therefore $L = 2\sqrt{R^2 - D^2}$.

Converting the long-chord condition

Using $L=2\sqrt{(R^2-D^2)}$, express the strict condition $L>\sqrt{3}R$ as an inequality for D .

The perpendicular from the centre bisects the chord, giving

$$(L/2)^2 + D^2 = R^2, \quad L = 2\sqrt{R^2 - D^2}.$$

Now apply the strict threshold:

$$2\sqrt{R^2 - D^2} > \sqrt{3} R.$$

Both sides are nonnegative, so squaring is valid:

$$4(R^2 - D^2) > 3R^2 \Leftrightarrow D^2 < R^2/4.$$

Since $D \geq 0$, the event is $D < R/2$.

A numerical geometry check

For a circle of radius $R=10$ cm and $L=2\sqrt{(R^2-D^2)}$, calculate L when $D=3$ cm and when $D=6$ cm. Compare both lengths with $\sqrt{3}R$.

At $D = 3$ cm: $L = 2\sqrt{100 - 9} = 2\sqrt{91} \approx 19.08$ cm.

At $D = 6$ cm: $L = 2\sqrt{100 - 36} = 16$ cm.

The threshold is $10\sqrt{3} \approx 17.32$ cm.

The first chord is longer than the threshold; the second is shorter. This agrees with the test $D < 5$ cm.

Coordinates that specify a chord

For a circle with radius R , let $t=D/R \in (0,1)$ and $\phi \in [0,2\pi)$ be the direction of the chord midpoint. Let $C(\phi,t)$ denote the chord through the point $R t(\cos\phi, \sin\phi)$, perpendicular to that radius.

Let $t = D/R$ and let ϕ be the direction of the midpoint from the centre.

Place the midpoint at distance Rt in direction ϕ , then draw the perpendicular chord.

Call this construction $C(\phi, t)$.

We use $0 < t < 1$ and $0 \leq \phi < 2\pi$. Diameters and zero-length chords have probability zero under the three models.

Recovering the chord coordinates

For a chord with midpoint m in a circle of radius R , define $t=\|m\|/R$ and $\phi=\text{Arg}(m)$ in $[0,2\pi)$. Recover (ϕ,t) from the chord.

Measure its normalized midpoint distance and direction:

$$t = \frac{\|m\|}{R}, \quad \phi = \text{Arg}(m) \in [0, 2\pi).$$

Thus

$$C^{-1}(c) = (\phi, t).$$

This is an inverse function on the stated chord space. The supplement gives the endpoint formula and verifies both inverse identities.

Model B: uniform radial distance

Sample D uniformly on $(0,R)$ and an independent uniform direction ϕ . For $L=2\sqrt{(R^2-D^2)}$, find the probability of $L>\sqrt{3}R$.

The event is $D < R/2$.

The favourable radial interval has length $R/2$, out of total length R :

$$P_B(L > \sqrt{3} R) = \frac{R/2}{R} = \frac{1}{2}.$$

This model is uniform in radial distance.

Model B: sampling map and inverse

Let U, V be independent uniform variables on $(0,1)$ and $[0,1)$, and set $t=U$, $\phi=2\pi V$. Using $C(\phi, t)$, write the chord, recover (U, V) , and express its length.

$$t = U, \quad \phi = 2\pi V, \quad c = C(2\pi V, U).$$

From the sampled chord, recover

$$U = t, \quad V = \phi/(2\pi).$$

The length is $L_B = 2R\sqrt{1 - U^2}$, and the long-chord event is $U < 1/2$.

Model C: a midpoint uniform in disk area

Choose the chord midpoint uniformly by area in a disk of radius R . The long-chord event is $D < R/2$. Find the probability.

Long chords have midpoints within radius $R/2$.

Uniform area sampling gives

$$P_C(L > \sqrt{3} R) = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}.$$

The smaller disk occupies a quarter of the total area.

Why uniform area uses a square root

For a midpoint sampled uniformly by area in a radius- R disk and U uniform on $(0,1)$, derive $P(D < r)$ and the corresponding transformation of U .

$$P(D < r) = \frac{\pi r^2}{\pi R^2} = \left(\frac{r}{R}\right)^2.$$

The transformation $D = R\sqrt{U}$ has exactly this property:

$$P(R\sqrt{U} < r) = P\left(U < (r/R)^2\right) = (r/R)^2.$$

Using $D = RU$ instead would give uniform radial distance.

Model C: sampling map and inverse

For uniform disk area, let $t=\sqrt{U}$ and $\phi=2\pi V$, where U,V are independent uniform variables on $(0,1)$ and $[0,1)$. Using $C(\phi,t)$, write the chord, its inverse and its length.

$$t = \sqrt{U} , \quad \phi = 2\pi V, \quad c = C(2\pi V, \sqrt{U}).$$

Inverse: $U = t^2$, $V = \phi/(2\pi)$.

The length is $L_C = 2R\sqrt{1-U}$.

The long-chord event is $U < 1/4$, which has probability $1/4$.

Practice: a smaller midpoint region

For the same disk of radius R , find $P(D < R/3)$ under (i) a uniform radial distance and (ii) a midpoint chosen uniformly by disk area.

Model B, uniform radial distance:

$$P_B(D < R/3) = \frac{R/3}{R} = \frac{1}{3}.$$

Model C, uniform area:

$$P_C(D < R/3) = \frac{\pi(R/3)^2}{\pi R^2} = \frac{1}{9}.$$

The region is identical. The two sampling distributions assign it different probabilities.

Model A: two uniform endpoints

Choose independent uniform endpoint angles A_1, A_2 on $[0, 2\pi)$, and let $\Theta = (A_2 - A_1) \bmod 2\pi$. Express the chord length using Θ .

After the first angle is chosen, rotational symmetry leaves Θ uniform on $[0, 2\pi)$.

Bisect the isosceles triangle formed by the centre and the two endpoints. Then

$$L/2 = R\sin(\Theta/2), \quad L = 2R\sin(\Theta/2).$$

For $\Theta = \pi$, the endpoints are opposite and $L = 2R$.

Model A: the favourable angular interval

Let $\Theta = (A_2 - A_1) \bmod 2\pi$ be uniform on $[0, 2\pi)$ and $L = 2R \sin(\Theta/2)$. Solve the strict inequality $L > \sqrt{3}R$ for Θ .

$$\sin(\Theta/2) > \sqrt{3}/2.$$

Since $0 \leq \Theta/2 < \pi$,

$$\pi/3 < \Theta/2 < 2\pi/3.$$

Therefore $2\pi/3 < \Theta < 4\pi/3$, and

$$P_A(\text{long}) = \frac{4\pi/3 - 2\pi/3}{2\pi} = \frac{1}{3}.$$

Model A in midpoint coordinates

For a uniform relative angle Θ on $[0, 2\pi)$, define $\alpha = \min(\Theta, 2\pi - \Theta)$. Find the distribution of α and the chord midpoint distance $t = D/R$.

$$\alpha = \min(\Theta, 2\pi - \Theta), \quad 0 \leq \alpha \leq \pi.$$

The right triangle gives $t = D/R = \cos(\alpha/2)$.

For $0 \leq a \leq \pi$, the angular intervals producing $\alpha \leq a$ have total length $2a$:

$$P(\alpha \leq a) = \frac{2a}{2\pi} = \frac{a}{\pi}.$$

Thus α is uniform on $[0, \pi]$.

Model A: sampling map and inverse

For independent uniform endpoint angles, put $U=\alpha/\pi$, where $\alpha=\min(\Theta, 2\pi-\Theta)$, and let $V=\phi/(2\pi)$ be the normalized midpoint direction. With $t=\cos(\alpha/2)$, find the chord map and its inverse.

$$t = \cos(\pi U/2), \quad c = C(2\pi V, \cos(\pi U/2)).$$

Inverse:

$$U = \frac{2}{\pi} \arccos t, \quad V = \phi/(2\pi).$$

This generates the same distribution of unordered chords as independent uniform endpoints. The supplement proves the independence of the midpoint direction and the smaller separation.

Model A: the length random variable

Under independent uniform endpoint angles, Θ is uniform on $[0, 2\pi)$, $\alpha = \min(\Theta, 2\pi - \Theta)$, $U = \alpha/\pi$ and $t = \cos(\alpha/2)$. Express $L = 2R\sqrt{1-t^2}$ in terms of U ; find the condition for $L > \sqrt{3}R$.

$$L_A = 2R\sqrt{1 - \cos^2(\pi U/2)} = 2R\sin(\pi U/2).$$

The sine is increasing for $0 < U < 1$, so

$$L_A > \sqrt{3} R \Leftrightarrow U > 2/3.$$

Therefore $P_A(\text{long}) = 1 - 2/3 = 1/3$.

Practice: recovering the source coordinates

For $t=1/2$ and midpoint angle $\phi=\pi$, recover $v=\phi/(2\pi)$ and u under Model A, $t=U$; Model B, $t=U$; and Model C, $t=\sqrt{U}$. Is the chord longer than $\sqrt{3}R$?

In every model, $v = \phi/(2\pi) = 1/2$.

Model A: $u = (2/\pi)\arccos(1/2) = 2/3$.

Model B: $u = t = 1/2$.

Model C: $u = t^2 = 1/4$.

The chord length equals $\sqrt{3} R$, so it is outside the strict long-chord event in all three models.

Comparing the three sampling distributions

Summarize the event calculations using the same uniform source U .

Sampling rule	Normalized distance t	Long-chord condition	Probability
A: uniform endpoints	$\cos(\pi U/2)$	$U > 2/3$	1/3
B: uniform radial distance	U	$U < 1/2$	1/2
C: uniform midpoint area	$\sqrt{U}$	$U < 1/4$	1/4

Each result uses a specified random variable and a specified sampling distribution.

Resolution of Bertrand's paradox

Why are the three probabilities compatible?

The geometric event is the same: all chords whose midpoint distance is less than $R/2$.

The sampling rules assign different probabilities to regions of the chord space.

“Random chord” does not select one of these probability distributions.

Compare the sampling experiments · Inspect the maps and inverses

Changing coordinates in one fixed model

Keep Model B, where $T = D/R$ is uniform on $(0, 1)$. Write $W = T^2$. What is $P(W < 1/4)$?

$$W < 1/4 \Leftrightarrow T^2 < 1/4 \Leftrightarrow T < 1/2.$$

Hence $P_B(W < 1/4) = 1/2$.

W is not uniform: in general, $P_B(W < w) = \sqrt{w}$ for $0 < w < 1$.

Declaring W uniform would change the sampling model to Model C.